

当然，以下是你所提供文档的**全英准确翻译**。为确保准确性，所有专业术语和结构均严格参考原文，部分术语采用国际标准写法（如“功能点”写为Function Point/FP），并在适当处用括号补充原文词汇以便对照。

Appendix 1:

Course Project Group: Smart Curtain Wall Project Team

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Course Project Title: Smart Curtain Wall — Stone Curtain Wall Stain Recognition System

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Part I: Software Size Measurement

1. Experimental Purpose and Principles

Experimental Purpose:

- Understand the basic concepts of software cost estimation, and practice the software size measurement process
- Based on the course project design documents (requirements analysis and design specifications), measure the software project size using the Function Point Analysis (FPA) method (in Function Points - FP)
- Provide basic data for subsequent cost estimation

Experimental Principles:

Software size measurement is the foundation of software cost estimation. FPA measures scale from the perspective of software functional characteristics. Mainstream methods include IFPUG, NESMA, GB/T 36964, etc.

This experiment adopts the GB/T 36964 method (equivalent to NESMA detailed method), including:

- Identifying data functions: Internal Logical Files (ILF) and External Interface Files (EIF)
- Identifying transactional functions: External Inputs (EI), External Outputs (EO), External Queries (EQ)
- Calculating Unadjusted Function Point (UFP): Based on function type and complexity (Low, Average, High)
- Applying Value Adjustment Factor (VAF): Adjusting UFP based on 14 General System Characteristics (GSC) to get Adjusted Function Point (AFP)

Core Concepts:

- **Internal Logical File (ILF):** User-recognizable logical data groups, e.g., user data
- **External Interface File (EIF):** Shared data between systems, e.g., OSS image storage
- **External Input (EI):** Data crossing the system boundary, e.g., image upload
- **External Output (EO):** Process results delivered to users, e.g., stain report

- **External Query (EQ):** Input-output combination queries, e.g., historical records query

2. Experimental Content and Steps

2.1 Identify System Boundaries

- **System Boundary:** Smart Curtain Wall Stain Recognition System (independent subsystem, Docker integrated into main system)
- **External Users:** Building maintenance companies, property management companies, architectural design companies
- **External Systems:**
 - User Management System (handles login, permissions) - **FTR=1**
 - OSS Database (stores images) - **FTR=1**
 - MySQL Database (stores history) - **FTR=1**
 - External AI Model (e.g., YOLO stain detection model) - **FTR=1**
- **System Scope:**
 - Core Functions: Image upload and stain recognition, history viewing, user management
 - Non-functional Requirements: Performance, security, compatibility

2.2 Function Point Analysis

Based on GB/T 36964 (NESMA detailed method), identify functions and determine complexity (referencing NESMA rules: ILF/EIF based on RET/DET, EI/EO/EQ based on FTR/DET).

Function Type	Description	RET	DET	FTR	Complexity
Data Functions					
ILF (Internal Logical Files)					
1. UserData	User info (username, password, permissions)	1	5	-	Low
2. HistoryData	Historical records (inspection time, image URL, result)	1	7	-	Low
3. StainResultData	Stain detection result (location, type, extent)	1	6	-	Low
Subtotal: ILF=3					RET=1 (single record type), DET<20→Low
EIF (External Interface Files)					
1. OSSImageStorage	Images stored in OSS	1	2	-	Low
2. ExternalAIModel	YOLO model output	1	3	-	Low

Function Type	Description	RET	DET	FTR	Complexity
Subtotal: EIF=2					RET=1, DET<5→Low
Transactional Functions					
EI (External Input)					
1. UploadImage	Upload image (JPG/PNG, ≤10MB)	-	5	3	High
2. Login	User Login (username, password)	-	4	1	Low
3. UpdateSettings	Change settings (e.g. change avatar)	-	4	1	Low
4. DeleteHistory	Delete history record	-	3	1	Low
Subtotal: EI=4	EI1:High (DET>4&FTR>1) EI2-4:Low (DET<4)				
EO (External Output)					
1. DisplayResult	Show stain detection result (labeled image, stain percentage)	-	8	2	High
2. GenerateReport	Generate historical report (PDF/CSV)	-	6	1	Average
3. ExportData	Export stain statistic data	-	5	2	Average
Subtotal: EO=3	EO1:High (DET>5); EO2-3:Average (DET=5-6)				
EQ (External Query)					
1. QueryHistory	Query history (filter by time)	-	5	1	Low
2. StainTypeStats	Stain type statistics and analysis	-	4	2	Low
Subtotal: EQ=2	DET<6→Low complexity				

2.3 Calculate Unadjusted Function Point (UFP)

According to NESMA detailed method:

Complexity Weights:

- ILF: Low=7, Average=10, High=15
- EIF: Low=5, Average=7, High=10
- EI: Low=3, Average=4, High=6
- EO: Low=4, Average=5, High=7
- EQ: Low=3, Average=4, High=6

Function Type	No. Low	No. Average	No. High	Calculation	Subtotal
ILF	3	0	0	$3 \times 7 = 21$	21
EIF	2	0	0	$2 \times 5 = 10$	10
EI	3 (Login, Update, Delete)	0	1 (UploadImage)	$(3 \times 3) + (1 \times 6) = 9 + 6 = 15$	15
EO	0	2 (Generate, Export)	1 (Display)	$(2 \times 5) + (1 \times 7) = 10 + 7 = 17$	17
EQ	2 (Query, Stats)	0	0	$2 \times 3 = 6$	6
Total UFP					69

2.4 Calculate Adjusted Function Point (AFP)

2.4.1 Value Adjustment Factor (VAF) Calculation Based on 14 General System Characteristics, score 0-5 points:

GSC No.	Description	Score (0-5)	Remarks
1	Reliable backup and recovery	3	Basic backup
2	Data communication requirement	5	Fully encrypted communication
3	Degree of distributed processing	2	No distributed architecture
4	Performance requirement	4	High real-time requirement
5	Operating environment	3	Standard environment
6	Degree of online data input	5	Mainly online operation
7	Multi-screen input	2	No multi-screen interaction
8	Online main file update	3	Moderate frequency
9	Complex input/output/query	4	Complex output processing
10	Complex data processing	4	Complex image analysis
11	Code reusability	3	Moderate reusability
12	Conversion and installation design	2	Common deployment
13	Multiple organization installation	1	No multi-org requirement
14	Ease-of-use requirement	4	High UX

Total Score (ΣGSC): 45

VAF Calculation:

$$VAF = 0.65 + \left(\frac{\sum GSC}{100}\right) = 0.65 + \left(\frac{45}{100}\right) = 0.65 + 0.45 = 1.10$$

AFP Calculation:

$$AFP = UFP \times VAF = 69 \times 1.10 = 75.90 \text{ FP}$$

(Rounded to two decimal places as per GB/T 36964 standard)

2.5 Using Multiple Methods for Verification

2.5.1 NESMA Indicative Method

- Data functions: ILF + EIF = 3 + 2 = 5
- Transactional functions: EI + EO + EQ = 4 + 3 + 2 = 9
- UFP = (35 × data functions) + (15 × transactional functions) = (35 × 5) + (15 × 9) = 175 + 135 = 310
- AFP = 310 × 1.10 = **341.00 FP**

2.5.2 NESMA Estimated Method

Function Type	Count	Weight	Calculation
ILF	3	10	30
EIF	2	7	14
EI	4	4	16
EO	3	5	15
EQ	2	4	8
UFP			83

AFP = 83 × 1.10 = **91.30 FP**

2.5.3 IFPUG Method Verification

Function Type	Low	Average	High	Subtotal
ILF	3 × 7 = 21	0	0	21
EIF	2 × 5 = 10	0	0	10
EI	3 × 3 = 9	0	1 × 6 = 6	15
EO	0	2 × 5 = 10	1 × 7 = 7	17
EQ	2 × 3 = 6	0	0	6

Function Type	Low	Average	High	Subtotal
UFP				69

$AFP = 69 \times 1.10 = 75.90 \text{ FP}$

2.5.4 Method Comparison

Method	UFP	VAF	AFP	Difference
NESMA Detailed	69	1.10	75.90	Baseline
NESMA Indicative	310	1.10	341.00	+349.27%
NESMA Estimated	83	1.10	91.30	+20.29%
IFPUG	69	1.10	75.90	0%

Conclusion:

- Detailed and IFPUG results are consistent
- Estimated method variance from detailed is 20.29%, acceptable (<30%)
- Indicative method difference is too large (349.27%), not suitable for this project

Part II: Software Cost Estimation

1. Experimental Purpose and Principles

Experimental Purpose

- Estimate software cost based on software size (AFP)
- Practice the cost estimation process: select productivity benchmark, convert to calendar time, and calculate cost (RMB)
- Verify results via another function point analysis method

Experimental Principles

Software cost estimation formula: $\text{Total Cost} = \text{Total Effort} \times \text{Hourly Rate}$

Where:

- $\text{Total Effort (HH)} = \text{AFP} \times \text{Productivity (HH/FP)}$
- $\text{Hourly Rate (RMB/HH)} = \text{Labor Rate (RMB/HM)} / \text{Work Hours per Month (HH/HM)}$
- $\text{Calendar Time (months)} = \text{Total Effort (HH)} / \text{Work Hours per Month}$

2. Experimental Steps & Results

2.1 Obtain Software Size

$AFP = 75.90 \text{ FP}$

2.2 Select Productivity Benchmark

- **Source:** Beijing Software Cost Evaluation Technology Innovation Alliance (BSCEA) 2023 industry benchmark
- **Project Type:** Image recognition system (refer to BSCEA classification)
- **Team Capability:** Medium (student team)
- **Technical Complexity:** High (AI integration)
- **Productivity:** P50 (median) = **8.25 HH/FP** (from BSCEA 2023 Report)

2.3 Convert to Calendar Time

Calculations:

- Total Effort (HH) = AFP × Productivity = 75.90 FP × 8.25 HH/FP = **626.175 HH**
- Work hours/month = 22.5 days/month × 8 hours/day = **180 HH/month**
- Calendar Time (months) = Total Effort / Work hours per month = 626.175 / 180 ≈ **3.48 months**

2.4 Obtain Software Cost

Detailed Calculation:

- **Labor Rate Source:** BSCEA 2023 data (average for software dev in Tier 1 cities)
- **Labor Rate:** 35,800 RMB/person·month (HM)
- **Hourly Rate** = Labor Rate / Work hours per month = 35,800 / 180 ≈ 198.89 RMB/HH
- **Total Cost (RMB)** = Total Effort × Hourly Rate = 626.175 × 198.89 ≈ **124,500 RMB**

Cost Breakdown:

Project Phase	Workload Share	Effort (HH)	Cost (RMB)
Requirements Analysis	15%	93.93	18,680
Design	20%	125.24	24,900
Coding/Implementation	35%	219.16	43,580
Testing & Validation	20%	125.24	24,900
Deployment & Maintenance	10%	62.62	12,450
Total	100%	626.175	124,500

2.5 Validate Cost Estimation Results (IFPUG Method)

Using IFPUG AFP from Part I (75.90 FP):

- Total Effort = 75.90 × 8.25 = 626.175 HH
- Hourly Rate = 198.89 RMB/HH (same as above)
- Total Cost = 626.175 × 198.89 ≈ 124,500 RMB

Comparison:

- GB/T 36964 cost = 124,500 RMB
- IFPUG cost = 124,500 RMB
- Difference = 0% (< 20%), validation passed

2.6 Experimental Summary

Indicator	Value	Unit	Remarks
Functional Size (AFP)	75.90	FP	GB/T 36964 method
Productivity Benchmark	8.25	HH/FP	BSCEA 2023
Total Effort	626.18	HH	AFP × Productivity
Work Hours/Month	180	HH/HM	22.5 days × 8 hours
Calendar Time	3.48	months	Effort / Work Hour/Month
Labor Rate	35,800	RMB/HM	BSCEA 2023
Hourly Rate	198.89	RMB/HH	Labor Rate / Work hours/month
Total Cost	124,500	RMB	Rounded to nearest hundred

Project Management Application:

1. **Size Baseline:** 75.90 FP as project baseline; reevaluation triggered if >10% change
2. **Progress Control:**
 - Design Stage (20%): Focus on architecture design & technical scheme
 - Coding (35%): Agile iterations, deliver usable version every 2 weeks
3. **Cost Control:**
 - AI model development (40% of cost): Separate budget monitoring
 - 10% contingency reserve for risk mitigation
4. **Resource Allocation:**
 - Dev Team: 4 people (2 frontend + 1 backend + 1 algorithm)
 - Testing: Dedicated testers from other teams for test phase

3. Thinking

Question: If the results of two software cost estimation methods differ by more than 20%, how would you solve it?

Answer:

Cause of Difference	Likelihood	Detection	Solution
Omission in function identification	High	Compare requirement and function matrix	Fill missing ILF/EIF/EI/EO/EQ
Complexity estimation bias	Medium	Expert review meeting	Invite 3 experts for independent scoring and mean value

Cause of Difference	Likelihood	Detection	Solution
Inappropriate productivity selection	Medium	Benchmark comparison	Refer to ISBSG, QSM, etc.
Parameter assumption error	Low	Sensitivity analysis	Test impact with productivity at 6.5–9.0 HH/FP
Project type misclassification	Low	Reclassify	Use BSCEA project classifier tool for review

Conclusion: In this experiment, both methods (GB/T 36964 and IFPUG) gave identical results (0% difference), validating the reliability of the estimation.

4. Learning Experiences

Group Members' Reflections:

- **Shi-Yu Luo:** This experiment deepened my understanding of the core role of function point analysis in project management. Through accurately defining system boundaries and detailed classification of function components, I mastered how to translate user requirements into measurable function points. The VAF rating step was inspiring and showed me the significant impact of non-functional requirements. I plan to further practice the GB/T 36964 standard to improve measurement accuracy.
- **Ya-Lin Bi:** Software cost estimation is a systematic project involving size measurement, productivity selection, and cost calculation. I focused on the use of the BSCEA database and found that AI-oriented projects (like our stain recognition system) require adjustment of productivity parameters. These experiences are crucial for future involvement in real project budgeting.
- **Bao-Ying Fu:** By comparing NESMA detailed, estimated, and indicative methods, I understood how method choice affects results. The detailed method, though time-consuming, is the most reliable, while the estimated method is more efficient in uncertain requirements. Phase-wise workload distribution in cost calculation is very practical—especially the finding that AI development accounts for 40%, reminding us to closely manage algorithm development risks.
- **Yu-Jie Wen:** The most valuable part was mapping function identification to the actual system. By repeatedly cross-checking requirements and design docs, we found initially omitted features like "stain type statistics" and "data export." This made me realize function point analysis is not just for calculation but also for ensuring requirement completeness. Also, the zero-difference result in cost verification proved the rigor of our process.

5. References

- Experiment Instruction (**Appendix 1**)
- Industry Standards:
 - GB/T 36964-2018 "Software Engineering — Software Development Cost Assessment Specification"
 - ISO/IEC 20926:2009 (IFPUG FSM Method)
 - ISO/IEC 24570:2018 (NESMA Method)
- Data:
 - Beijing Software Cost Evaluation Technology Innovation Alliance (BSCEA), "2023 Software Industry Benchmark Data Report" (<https://www.bscea.org/>)

- International Software Benchmarking Standards Group (ISBSG) 20th Database
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